Royal Crest Academy
SCIENCE · INNOVATION · SKILL

FREE LEARNER GUIDE

Physics *study guide*

Physics explains why things move, how energy is transferred and how waves, electricity and atoms behave, from a bicycle to a satellite.

Big ideas

Forces change motion

Newton's laws link force, mass and acceleration (F = ma). Balanced forces keep the velocity constant; an unbalanced force makes an object accelerate.

Energy is conserved

Energy is never created or destroyed, only transferred between stores, although some is always dissipated as heat.

Waves carry energy

Sound, light and radio are waves. They transfer energy without transferring matter, and v = fλ links their speed, frequency and wavelength.

Topic checklist

tick each topic when you can explain it

O LEVEL

- [ ] Measurements and units
- [ ] Speed, velocity and acceleration
- [ ] Forces, mass and weight
- [ ] Turning effects and stability
- [ ] Work, energy and power
- [ ] Pressure
- [ ] Thermal physics
- [ ] Waves and sound
- [ ] Light: reflection and refraction
- [ ] Electricity and circuits
- [ ] Magnetism and electromagnetism
- [ ] Radioactivity

A LEVEL

- [ ] Physical quantities and units
- [ ] Kinematics and dynamics
- [ ] Forces, density and pressure
- [ ] Work, energy and power
- [ ] Deformation of solids
- [ ] Waves and superposition
- [ ] Electricity and d.c. circuits
- [ ] Electric and magnetic fields
- [ ] Circular motion and gravitation
- [ ] Thermal physics and ideal gases
- [ ] Oscillations
- [ ] Quantum and nuclear physics

Key terms

Velocity	Speed in a stated direction.
Acceleration	The rate of change of velocity, measured in m/s².
Force	A push or a pull, measured in newtons (N).
Power	The rate of doing work or transferring energy, measured in watts (1 W = 1 J/s).
Current	The rate of flow of electric charge, measured in amperes (A).
Resistance	How much a component opposes the current: R = V ÷ I, measured in ohms (Ω).
Refraction	The change in direction of a wave as it enters a new medium, because its speed changes.
Half-life	The time taken for half of the radioactive nuclei in a sample to decay.

In the laboratory

- Measuring with rulers, vernier callipers and micrometers
- Timing the swing of a pendulum
- Finding the density of solids and liquids
- Plotting current–voltage graphs
- Ray diagrams with mirrors and lenses
- Measuring specific heat capacity

Study smarter

1. Learn every quantity with its unit and symbol.
2. Show working: formula, substitution, answer, unit.
3. Draw graphs with labelled axes and a best-fit line.
4. Explain answers using physics key words.

Check yourself

1. A car travels 150 m in 10 s. What is its average speed?

2. What is the weight of a 5 kg mass on Earth? (g = 10 N/kg)

3. A 12 V supply drives a current of 2 A through a resistor. What is its resistance?

Answers 1. 15 m/s (speed = distance ÷ time = 150 ÷ 10). · 2. 50 N (W = mg = 5 × 10). · 3. 6 Ω (R = V ÷ I = 12 ÷ 2).

Where it leads: Electrical engineering, Mechanical engineering, Civil engineering, Telecommunications, Energy and power, Radiography, Aviation